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SINCE 2008
SKILL EDUCATION
SHAPING SKILLS FOR SCALING HIGHER...!!!

Title: Holistic Data Preparer (Final Project)
Duration: 8 Days (1 hour per Day)

🎯 Objective:

The purpose of this project is to test your **complete knowledge** of **Data Preprocessing and Feature Engineering**. You will go through the **end-to-end process** of preparing a dataset for machine learning: from **data understanding** to **data cleaning, imputation, outlier handling, encoding, scaling, transformations, and feature engineering**. The outcome will be a **fully processed dataset** ready for ML modeling.

📑 Problem Statement

You are hired as a **Junior Data Scientist** at a fintech company. The company has provided a **Customer Credit Risk dataset** collected from multiple sources (CSV, JSON, SQL tables, and an external API). Your manager wants you to perform **full-scale preprocessing and feature engineering** so that the dataset is clean, consistent, and suitable for building a **Machine Learning model** that predicts whether a customer is likely to default on a loan.

The dataset contains:

- **Demographics** (age, gender, region, education, employment type).
- **Financial details** (income, loan amount, loan purpose, repayment history, credit score).
- **Behavioral attributes** (transactions, spending habits, missed payments).
- **Target variable**: Loan Default → 0 (No), 1 (Yes).

📁 Dataset Structure: Customer Credit Risk Dataset (Generate from an AI Tool)

Field Name	Data Type	Description	Notes (For Exam Tasks)
customer_id	String / Int	Unique identifier for each customer	No missing values
age	Integer	Age of customer (years)	Missing values injected (for imputation tasks)
gender	Categorical	Gender: Male / Female / Other	Missing + category imbalance
region	Categorical	Region of residence: North / South / East / West	Good for One-Hot Encoding
education_level	Ordinal Cat.	Education: Primary / Secondary / Graduate / Post-Graduate	For Ordinal Encoding
employment_type	Categorical	Employment: Salaried / Self-Employed / Unemployed	Missing values for categorical imputation
annual_income	Float	Annual income (₹)	Outliers (very high incomes) + missing values
loan_amount	Float	Loan amount requested (₹)	Outliers + skewed distribution
loan_purpose	Categorical	Purpose: Home / Car / Education / Business / Other	For One-Hot Encoding
credit_score	Float	Credit score (300–850)	Outliers + missing values
repayment_history	Integer	Number of missed payments in last 12 months	For Binning / Outlier treatment
transaction_count	Integer	Total number of transactions in last 6 months	For K-Means Binning
spending_ratio	Float	Spending-to-Income ratio (%)	To test log/Box-Cox/Yeo-Johnson transforms
join_date	Date	Date customer joined bank	For handling date/time variables (extract Y/M/D/W)
default_flag	Binary Int	Target variable: 0 = No Default, 1 = Default	ML Problem Target

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🧩 How This Dataset Enables All Exam Tasks

- **Missing Value Imputation** → `age`, `annual_income`, `credit_score`, `employment_type` have injected missing values.
- **Simple Imputer (Num + Cat)** → `age`, `employment_type`.
- **Most Frequent Imputation** → `gender`.
- **Missing Indicator + Random Sample** → `annual_income`.
- **KNN Imputer & MICE** → Applied across `annual_income`, `loan_amount`, `credit_score`.
- **Outlier Detection** → `annual_income`, `loan_amount`, `credit_score`.
- **Z-score / IQR / Percentile / Winsorization** → All tested on numerical features.
- **Handling Mixed Variables** → `gender` (cat), `age` (num), `spending_ratio` (num skewed).
- **Date Handling** → `join_date` → extract Year, Month, Weekday.
- **Encoding** →
  - Ordinal Encoding → `education_level`.
  - Label Encoding → `gender`.
  - One-Hot Encoding → `region`, `loan_purpose`.

- **Numerical Encoding** → `repayment_history` and `transaction_count`.
- **Binning & Discretization** → `annual_income`, `repayment_history`.
- **Binarization** → flag `credit_score > 700`.
- **Quantile & K-Means Binning** → `transaction_count`.
- **Feature Scaling** →
  - Standardization → `annual_income`, `loan_amount`.
  - MinMax / MaxAbs / Robust Scaling → all numeric columns.

- **Feature Construction** →

  - New features like `Debt-to-Income ratio = loan_amount / annual_income`.

- **Transformations** →
  - Log, Reciprocal, Square root on `spending_ratio`.
  - Box-Cox & Yeo-Johnson on `loan_amount` and `annual_income`.

- **Column Transformer** → Apply different preprocessing to categorical vs numeric features in one pipeline.

📝 Tasks

Part A: Conceptual Foundation

1. Write short notes on:
   - What is Data Analysis?
   - How to Plan a Data Science Project.
   - How to Frame a Machine Learning Problem.

2. Explain **Tensors** and provide an **in-depth explanation** with NumPy examples.

Part B: Data Acquisition

3. Import datasets from multiple sources:
   - Load CSV files (main transactions dataset).
   - Parse JSON files (customer metadata).
   - Fetch records from SQL (loan repayment history).
   - Fetch data from a dummy API (external economic indicators).

Part C: Data Understanding & Cleaning

4. Explore the dataset using Pandas (`.info()`, `.describe()`).
5. Perform **Pandas Profiling** to generate a **data quality report**.
6. Handle missing data with:
   - Simple Imputer (numerical: mean/median).

   - Simple Imputer (categorical: most frequent).
   - Most Frequent Category Imputation.
   - Missing Indicator + Random Sample Imputation.
   - KNN Imputer (multivariate).
   - MICE Algorithm.
   - Complete Case Analysis (dropping rows/columns).

Part D: Outlier Handling

7. Detect and treat outliers using:
   - Z-score Method.
   - IQR Method.
   - Percentile Method.
   - Winsorization Technique.

Part E: Feature Engineering

8. Handle variable types:
   - Mixed Variables (numeric + categorical).
   - Date & Time variables → extract Year, Month, Day, Weekday.

9. Encoding categorical variables:
   - Ordinal Encoding (education levels).
   - Label Encoding (binary features).
   - One-Hot Encoding (regions, loan purpose).

10. Encoding numerical features:
    - Binning (discretize income into groups).
    - Binarization (flag if > threshold).
    - Quantile Binning.
    - K-Means Binning.

Part F: Feature Scaling

11. Apply multiple scaling methods:
    - Standardization (Z-score scaling).

    - Normalization.
    - Min-Max Scaling.
    - MaxAbs Scaling.
    - Robust Scaling.

Part G: Feature Construction & Transformation

12. Apply transformations:
    - `FunctionTransformer` → log transform, reciprocal, square root.
    - `PowerTransformer` → Box-Cox and Yeo-Johnson.
    - Column Transformer → apply different preprocessing steps to different columns.

13. Construct new features:
    - Debt-to-Income ratio.
    - Average monthly transactions.
    - Spending-to-Income ratio.

Part H: Final Deliverable

14. Provide a **final cleaned and transformed dataset**.
15. Write a **report summarizing**:
    - Missing value strategies used and their effectiveness.
    - Outlier handling results.
    - Encoding methods applied to categorical/numerical variables.
    - Scaling/transformations applied and why.
    - Newly engineered features and their usefulness.
    - Final dataset shape and readiness for ML modeling.

📌 Expected Outcome

By the end of this **mega project**, students will:

- Understand how to **plan and execute a complete data preprocessing workflow**.
- Perform **detailed data cleaning** using imputation and outlier handling.
- Apply **advanced encoding and scaling techniques**.
- Construct and transform features to improve ML readiness.

- Generate a **high-quality dataset** that can directly be used to build a machine learning model.

✅ Submission Guidelines

- Include practical implementation in a Jupyter Notebook and screenshots.
- Label all charts (if any) clearly and write short interpretations under each result.
- **GitHub Repository:**
  - Create a GitHub repository to host your project.
  - Upload your project files, including source code, project explanation video, and documentation to the repository.
  - Add a document (PDF) explaining theory concepts with definitions (if any).
  - Ensure that you provide a clear and descriptive README.md file.

Remember to follow the instructions provided professionally, make suitable assumptions wherever necessary, and avoid copying code or content from unauthorized sources. Good luck with your project work!

Transparent Marking Scheme (10 Marks)

Total Marks: 10 — distributed across 3 components as follows:

Component	Marks	Weight
A. Technical Completion of Project	5	50%
B. Recorded Video Explanation (Face + Screen)	3	30%
C. GitHub Repository with README.md	2	20%
TOTAL	**10**	**100%**

Recorded Video — Face + Screen (3 Marks)

Record a video showing your face (webcam) and the entire screen simultaneously. Walk through and explain every task.

Criterion	Marks	Excellent (Full)	Good (75%)	Average (50%)	Poor (25%)	Zero
B1	1.0	Clear audio and video; face visible via webcam picture-in-picture throughout	Clear audio; face mostly visible	Audio unclear; face partial	Very poor quality	No video
B2	1.5	Explains every task with reasoning	Explains most tasks with reasoning	Walks through without explaining why	Superficial narration	No explanation
B3	0.5	5–10 minutes; covers all tasks	4–5 min; covers most tasks	2–4 min; misses key tasks	Under 2 minutes	Not submitted

Subtotal B: 1.0 + 1.5 + 0.5 = 3.0 Marks

Video Requirements

- Format: MP4 or MOV (file size ≤ 500 MB)
- Face must be visible via webcam picture-in-picture overlay throughout the recording
- Screen must clearly show entire screen
- Duration: 5 to 10 minutes — shorter or longer will lose marks on B3
- Naming convention: PR_Final_YourName_GRID.mp4
- Upload to Google Drive (Anyone with link can view) or YouTube (Unlisted)
- Paste the video link into your GitHub README.md before submitting

Holistic Data Preparer (Final Project)
Data Preprocessing and Feature Engineering

BRING ON YOUR CODING ATTITUDE